

Mathematics for Designers – Individual Project Report

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Hyperbolic paraboloids in Furniture Design

Introduction (abstract) -

A hyperbolic paraboloid is a unique and beautiful shape. As the name suggests it is created via the combination of hyperbolic and parabolic curves. The shape is such that its sections parallel to one coordinate plane are hyperbolas, with its sections parallel to the other two coordinate planes being parabolas. The general equation for the shape is $(z = \frac{x^2}{a^2} - \frac{y^2}{b^2})$ or to simplify $(z = ax^2 - by^2)$. The general shape curves in opposite ways along its two principal axes, this causes the entire shape to have a negative Gaussian curvature (Gaussian curvature is in simple terms the measure of curvature. If it is positive the curve is convex-curves in same direction and if it is negative the curve has a saddle-like shape-curves in opposite directions, as seen in hyperbolic paraboloids). The beauty and strength of the hyperbolic paraboloid is the shape's double curvature, as when a load is applied the inherent shape allows the load to be distributed throughout its form providing exceptional rigidity. Furthermore, the shape allows for minimal material costs as structures can be created with thin sheets that span a large area without the need of supporting column structures. Due to the simplicity of the shape and it being a 'ruled surface' it can be created completely out of straight lines making production simple in many ways. The key characteristics of the shape are perfect for roofs and walls in architecture as they carry load in an efficient and aesthetic manner, however this report's objective is to translate its qualities into the design field. After analysis the project chosen is creation of a coffee table that integrates the hyperbolic paraboloid as its primary structure. The field being furniture design, a scaled down model via 3d printing for form exploration and a finalised product that involves unique materials will be done. By creating such a structure, it will help in cost efficiency and aesthetic pleasure of future coffee tables as it allows for new materials to be experimented with rather than traditional wooden tables with load bearing columns. This can add a premium feel to a product that has a relatively low production cost and will hopefully lead to more complex mathematical shapes and architectural structures being integrated into furniture design.

Mathematical concept(s)

A hyperbolic paraboloid is a quadratic structure meaning it is a three-dimensional surface that can be defined using a quadratic equation. The general equation for the quadratic shape is-

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

In the equation x and y are the variables, a and b are the real nonzero constants that determine the curvature along the coordinate axes. The negative Gaussian curvature at every point signifies an opposing curvature along orthogonal axes.

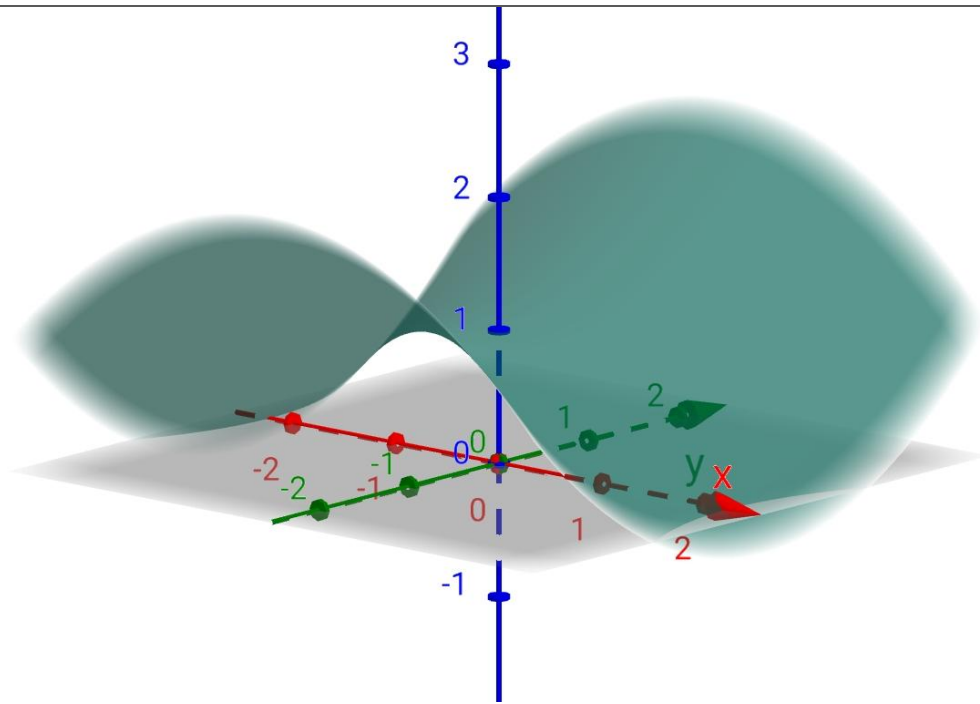


Fig 1.1

To further break down the constituents of the shape a brief introduction about parabolas and hyperbolas will be added.

Parabola

It is a conic section that is formed by the intersection of a plane and a cone. Parabolas are a u-shaped structure that are a representation of a quadratic equation. Their general formula is

$$y = ax^2 + bx + c$$

$$y = x^2$$

Parabolas have a wide range of applications in physics, engineering, and everyday life, such as design for satellite dishes and trajectory of projectiles. If a hyperbolic paraboloid we to be cut into half along all three of its coordinate axes, two of the axes would be represented as parabolas for the cross section, one being a convex parabola and the other a concave parabola.

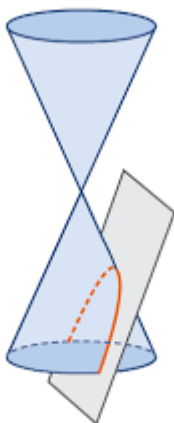


Fig 1.2

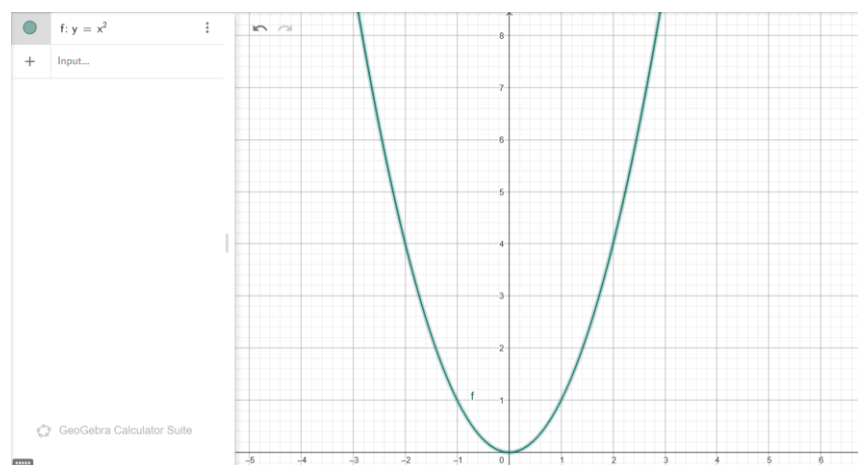


fig 1.3

Hyperbola

A hyperbola is a type of conic section that is an open curve with two branches that are mirror image of one another. It is defined as the set of all points where the absolute difference of two fixed points is constant. It is formed when a plane intersects a double cone at an angle the cuts through both halves of the cone. Its general formula is-

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Hyperbolas have a wide range of applications in industries such as they are used to model the shape of nuclear cooling towers and complex gear shapes. If a hyperbolic paraboloid we to be cut into half along all three of its coordinate axes, one of the axes would be represented as a hyperbola for the cross-section.

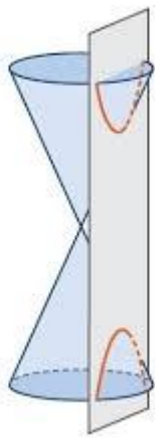


Fig 1.4

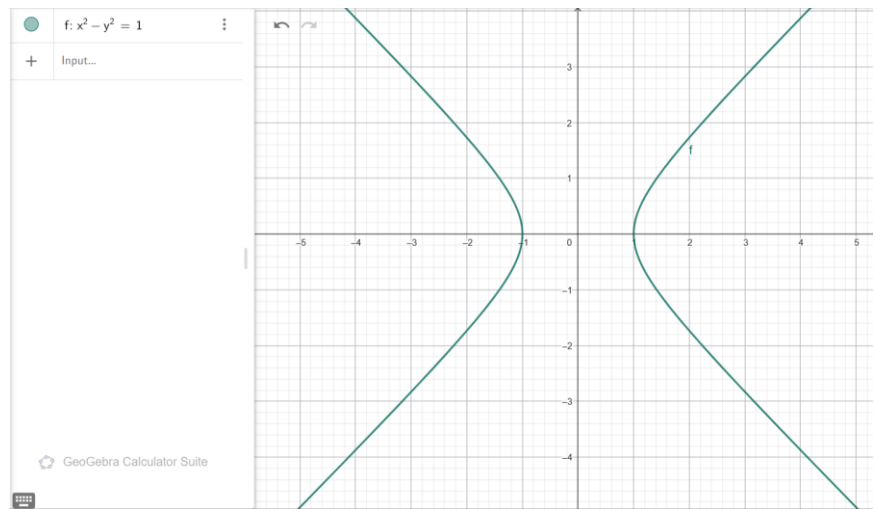


fig 1.5

Cross section analysis

The combination of the two shapes creates an elegant structure that is as both simple and complex. This is proven by ignoring a separate coordinate axis to find the equations of each. This is the general equation of the hyperbolic paraboloid used-

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = z$$

The following graphs are created via GeoGebra's 3d calculator mode. The following figures (Fig 1.6, Fig 1.7 and Fig 1.8) are all different views of the same quadratic shape. Each is taken from the perspective of a different plane such as Fig 1.6 illustrates the hyperbolic paraboloid from the y-z plane, while Fig 1.8 shows the shape from the x-y plane.

Cross-Sections

By replacing x with 0 we get a quadratic equation that implies that the y - z plane contains a parabola.

$$x = 0$$

$$-\frac{y^2}{b^2} = z$$

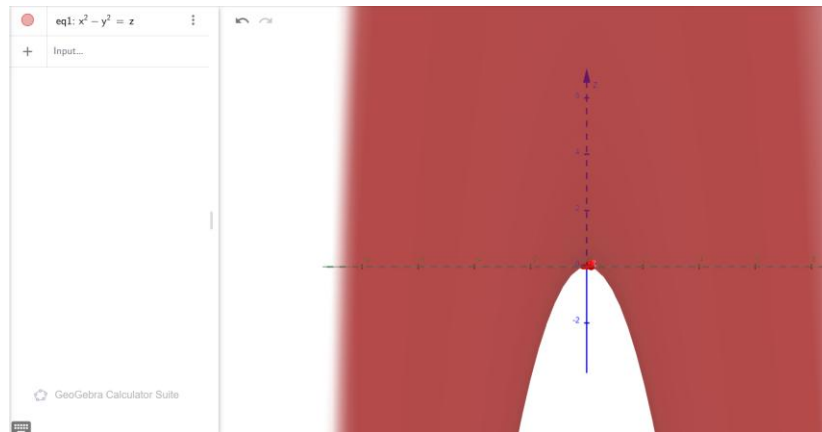


Fig1.6

By replacing y with 0 we get a quadratic equation that implies that the x - z plane contains a parabola.

$$y = 0$$

$$\frac{x^2}{a^2} = z$$

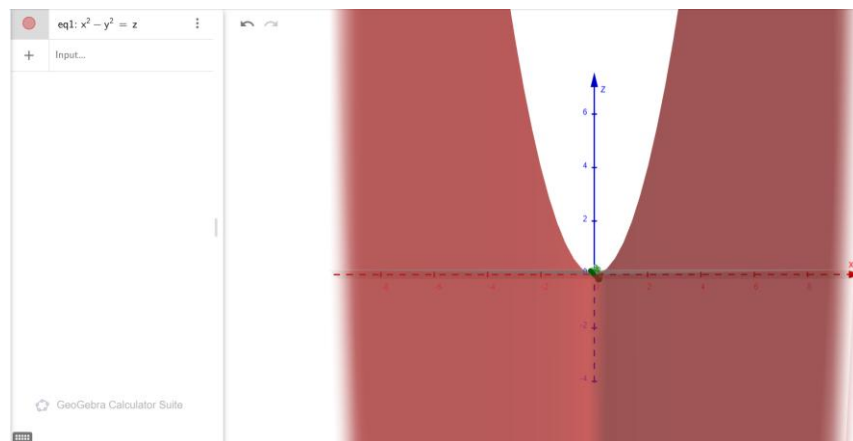


Fig1.7

By replacing z with 0 we get a quadratic equation that implies that the x - y plane contains a hyperbola.

$$z = 0$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

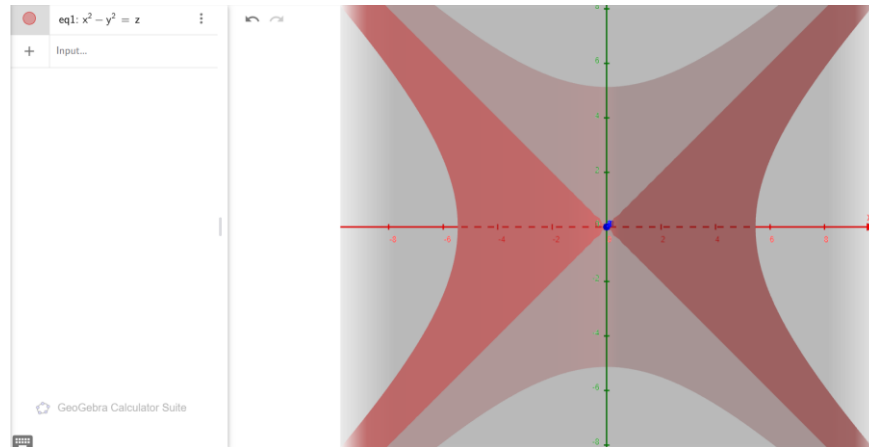


Fig1.8

Design context

The following product that will be designed is a double hyperbolic paraboloid coffee table that is inspired by the Bauhaus aesthetic. A summary of what will be done is a 3d model will be designed for the table using CAD software. Then in the future a material study to finalize the material in which the table will be created will be finalized. The strength and load carrying along with the form will all be calculated through mathematical means. A basic design will be created and elaborated on.

Coffee table study-

A summary and introduction to what is a coffee table and its defining characteristics- The coffee table is a staple in most houses and acts as the anchor to any living room area. It is a low table designed to be placed in front of any seating arrangement of choice and its primary purpose is to act as surface on which glasses, plates or any other item of the user's choice can be placed. Its defining characteristic is its low height, typically between 16 to 18 inches in height or 40 to 45 centimeters. The low profile ensures it doesn't obstruct the view of the people sitting around the piece and having conversation. Furthermore, besides the height of the coffee table the material, shape and design are open and vast as they can be made up of wood or metal or even stone, and the flat portion can be oval, square or even circle shaped. Thus, this allows for lots of variation and is the perfect base article of furniture for this project.

Bauhaus Aesthetic-

For the table the Bauhaus aesthetic is optimal as it focuses on and prioritizes functionality, simplicity and lack of ornamentation. Furthermore, the clean lines and geometric shapes fall in line with the topic at hand and allow for the use of modern materials such as steel and other unique polymers. The art style gives products an industrial and refined feel and is the perfect blend of form and function. A few examples of products are given below.

Coffee Table Design-

The table design involves two primary components- the base and the flat top. The base will consist of two hyperbolic paraboloids one is inverted and the other lies on top of it mimicking a mirror image. The base is connected to an ellipsis top that acts as the flattop for placing objects. The hollow cavity between the two hyperbolic paraboloids acts as a storage slot for keeping items that can be accessed in the future or momentary storage. The paraboloids are modeled by the equation-

($\frac{x^2}{5} - \frac{y^2}{5} + z = 1$) and the mirror image is the same structure rotated about the centroid of the object along the x-y plane.

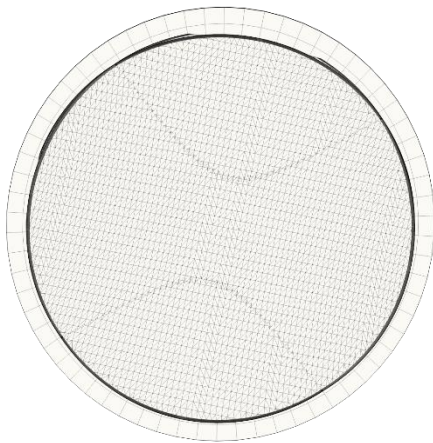


Fig 2.1

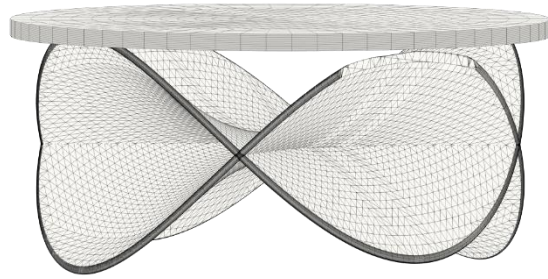


Fig 2.2

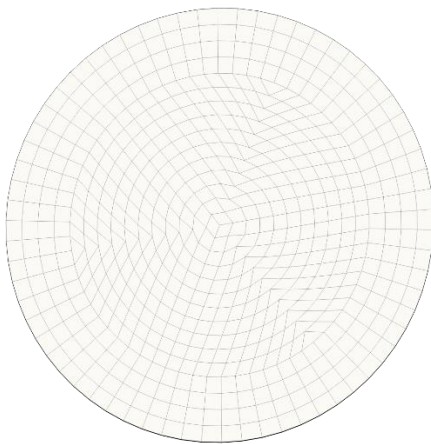


Fig 2.3

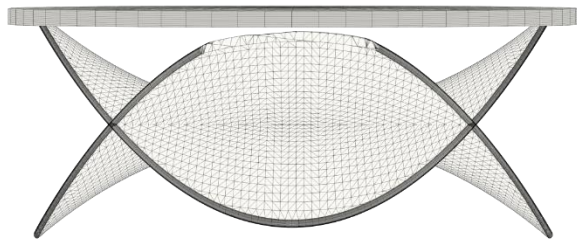


Fig 2.4

The following 3d model is created on SolidWorks modelling software. The form exploration for the table went through several stages of development to finalize on this prototype that creates maximum visual interest and stability. As shown within the different angles of models, the table displays the elements of hyperbolic paraboloid, A hyperbola on the bottom to top facing plane and parabolas on the front-end plane. The top is a simple circular plane that juxtaposes the complex structure underneath. The final

model is ideally metallic in nature, but the material can have variations and be changed in future prototypes. The final render is as shown below-

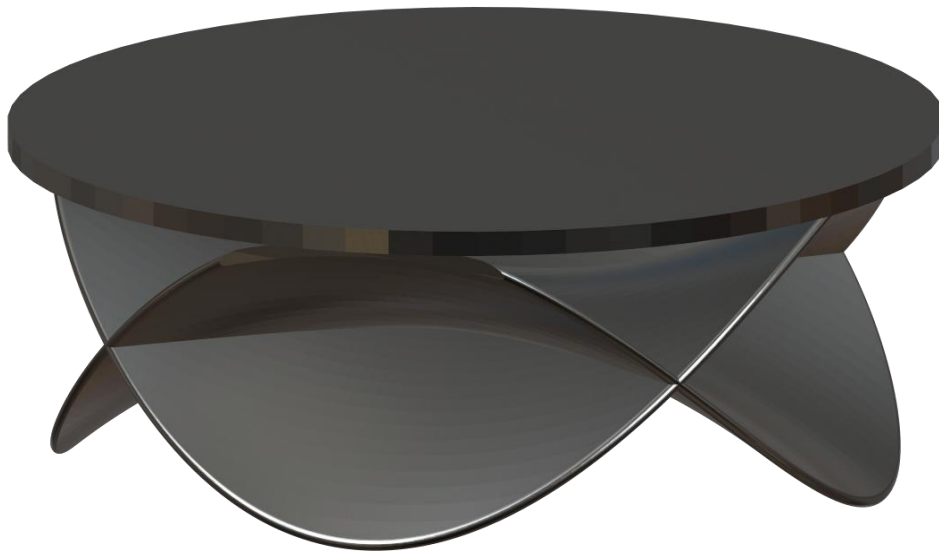


Fig 2.5

Observations

As seen in the Fig 2.5 there are four contact points with the flattop circular surface, this is optimal for load distribution as the weight of the objects places is evenly distributed along the form both hyperbolic paraboloids. Furthermore, the cavity formed underneath the topmost plane allows for additional storage of items and the shape allows for variety and ease in production methods. However, due to not enough research data the intersection between the two hyperbolic paraboloids can be cause for concern while manufacturing in terms of its load bearing qualities.

Conclusion

To conclude, while researching and developing the product for this research paper the main learning points were that the field of furniture design is vast and interesting, and it will be interesting to do more work in the field in the future. The key takeaways from this paper were the use of complex quadratic shapes in the field of design and architecture along with the fundamentals of how the structures are formed via 2d shapes and equations. In the future a deeper dive into material study will be done along with complex lattice structure development and more optimal form exploration.

During the ideation phase creation of a sound and unique product was extremely challenging at first, creating a balance between modern and interesting without the project looking like a gimmick. alongside, the issues and learning curve of 3d modelling there were other minute hurdles and challenges along the development process. However, in the end I learnt a lot about furniture and product design and was intrigued by use of mathematics in the design process.

Finally, for what makes the advantages of this design over a conventional coffee table is that this table with a hyperbolic paraboloid base can deliver more strength for less material, and a visually compelling form compared to a conventional table. For several users, it's a way to elevate what might otherwise be a mundane piece into something that communicates thoughtful geometry, structural efficiency, and acts as a conversation starter or a piece.

Tools used

GeoGebra, SolidWorks, Nomad Sculpt, Blender and MS Word.

Project link

https://jaacontractors.com/construction-engineering/guide-to-various-types-of-shell-structures/?utm_source

https://designhorizons.org/design-and-benefits-of-hyperbolic-paraboloid-roofs/?utm_source

<https://pdfs.semanticscholar.org/da88/1c8a39b475cf3cc0b76e39fc470786ff5137.pdf>

https://link.springer.com/chapter/10.1007/978-1-4757-0223-1_13

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